

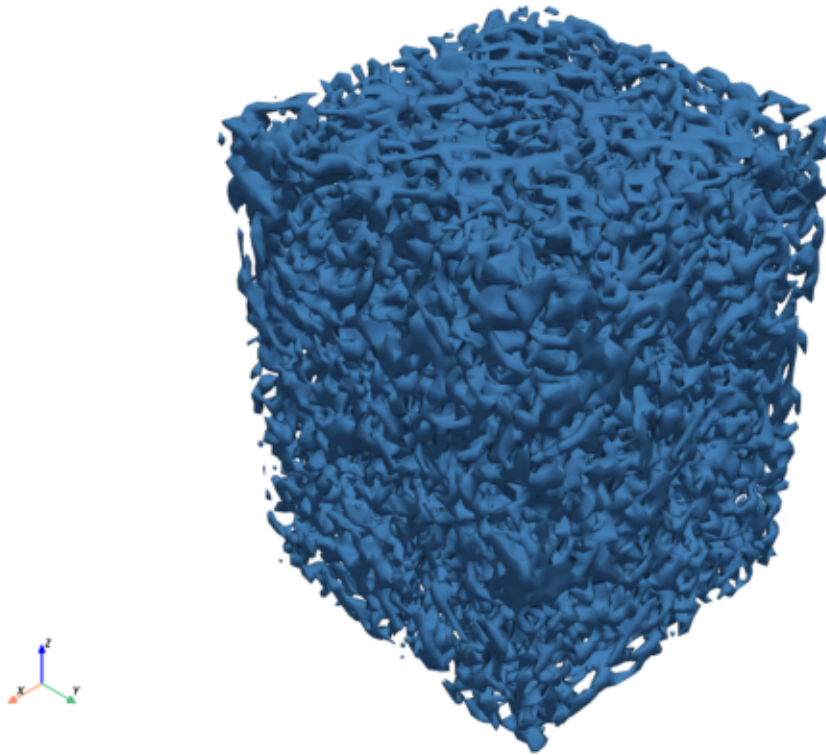
Bio-Inspired Heat Exchanger — Design Summary

Generated: 2026-06-16 20:43

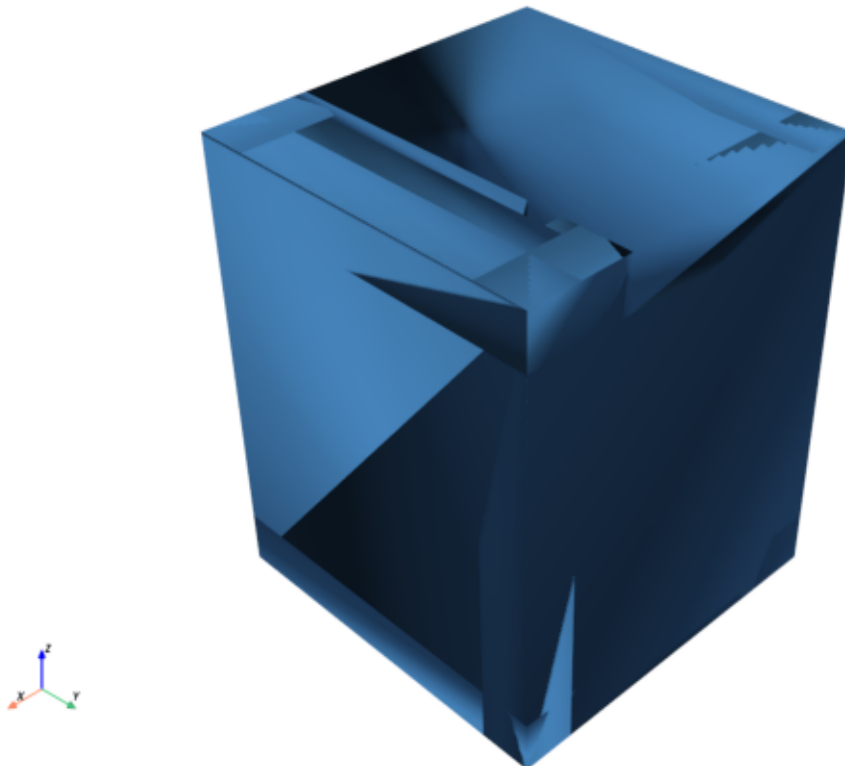
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encapsulation:
  length_mm..... 15.0
  width_mm..... 15.0
  height_mm..... 20.0
  wall_mm..... 3.0
  Dh_mm..... 10.957
  V_int_cm3..... 1.134
flow:
  v_in_m_s..... 0.1
  T_in_K..... 380.0
  p_out_Pa..... 0.0
  Re..... 1095.7
  Pr..... 6.852
sizing:
  Re_simulation..... 1095.7
  Re_operating..... 1000.0
  Nu_laminar..... 41.746
  Nu_turbulent..... 39.883
  target_Nu..... 10.0
  required_solidity..... 0.05
  wall_thickness_est_mm..... 0.1744
  h_W_m2K..... 2220.45
  lam_turb_factor..... 0.9554
lattice:
  tpms..... gyroid
  k0_rad_m..... 3490.6585039886586
  uc_mm..... 1.8
  phi..... 0.05
preview_geometry:
  solidity..... 0.1913
  mass_g..... 0.96
  surface_cm2..... 39.8
  n_triangles..... 217864
manufacturability:
  overhang_penalty..... 0.1075
  thickness_penalty..... 0.4518
  combined_penalty..... 0.5594
  est_wall_thickness_mm..... 0.1096
  min_feature_size_mm..... 0.2
  max_overhang_angle_deg..... 45.0
  printable..... False
```

Optimised Geometry

gyroid_surface_lattice.stl



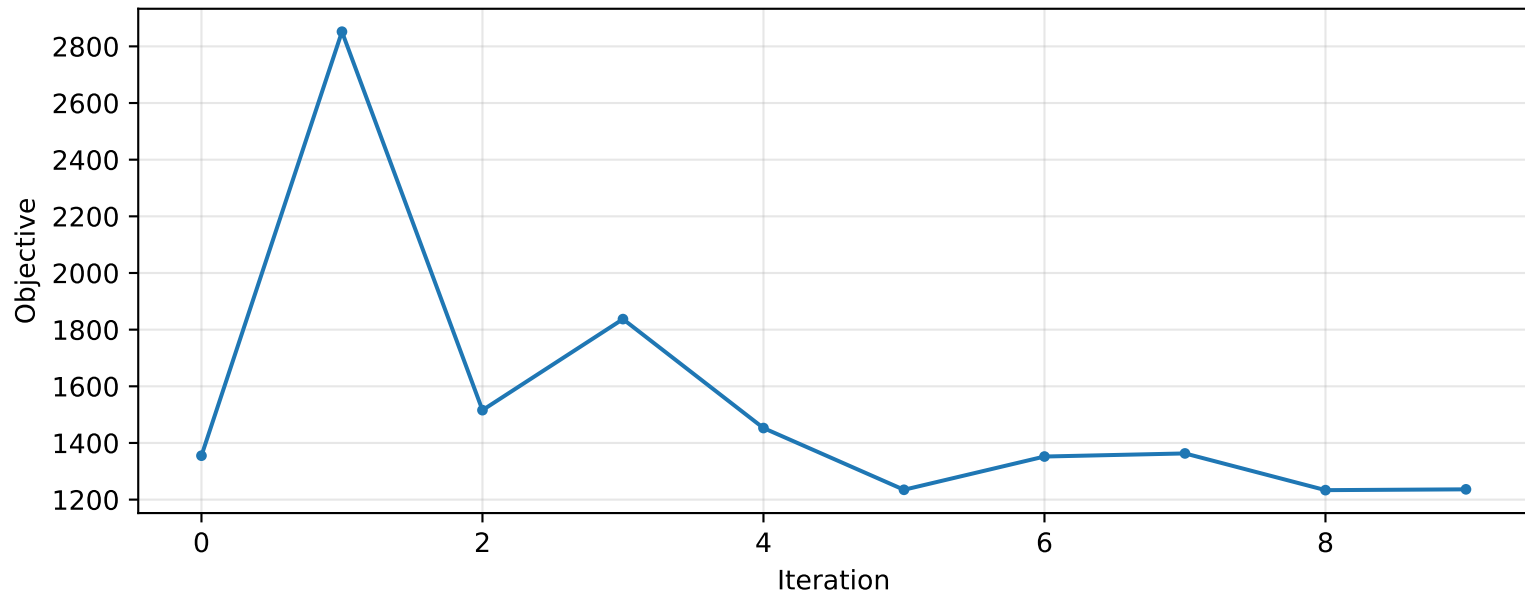
gyroid_surface_encap.stl



Optimisation History

iter	mode	J_meanT	J_obj	J_aug	DissPower	outletT	solid_frac	solid_g	flow_m3s	deltaP_m2s2	wall_thickness	g_meanT	pen_meanT	g_disspower	pen_disspower	mu_penalty	g_oh	pen_oh	mu_oh	grad_norm	delta_x	elapsed_s	alphaMax
1		23.87	1355	1356	1355	281.6	0.4005	4.871	2e-07	6.797	0.2	293.9	0	nan	0	1e+04	0.01497	1.147	8100	1436	nan	1088	4.444e+07
2		17.38	2852	2853	2852	278.4	0.4001	4.867	2e-07	14.29	0.2	287.4	0	nan	0	1e+04	0.0138	1.202	9720	801.3	18.84	311.7	4.444e+07
3		32.31	1516	1517	1516	286.2	0.401	4.878	2e-07	7.607	0.2	302.3	0	nan	0	1e+04	0.01125	1.126	1.166e+04	1124	18.84	313.9	4.444e+07
4		18.13	1837	1839	1837	279.6	0.4011	4.88	2e-07	9.218	0.2	288.1	0	nan	0	1e+04	0.01174	1.427	1.4e+04	333.5	17.75	321.7	4.444e+07
5		26.39	1453	1455	1453	282.8	0.4029	4.901	2e-07	7.29	0.2	296.4	0	nan	0	1e+04	0.01448	2.535	1.68e+04	1180	17.84	314.6	4.444e+07
6		25.09	1235	1237	1235	282.1	0.3994	4.858	2e-07	6.203	0.2	295.1	0	nan	0	1e+04	0.01141	1.858	2.016e+04	443.5	16.55	320.5	4.444e+07
7		23.27	1352	1355	1352	280.9	0.4001	4.867	2e-07	6.797	0.2	293.3	0	nan	0	1e+04	0.01305	2.806	2.419e+04	623.8	15.95	318.5	4.444e+07
8		23.72	1363	1367	1363	281.6	0.404	4.914	2e-07	6.846	0.2	293.7	0	nan	0	1e+04	0.01312	3.458	2.902e+04	782.1	14.82	322.1	4.444e+07
9		24.83	1233	1238	1233	282.4	0.402	4.89	2e-07	6.193	0.2	294.8	0	nan	0	1e+04	0.01393	4.398	3.483e+04	295.7	13.79	320.1	4.444e+07
10		27.51	1236	1242	1236	283.6	0.3989	4.852	2e-07	6.211	0.2	297.5	0	nan	0	1e+04	0.01442	5.591	4.179e+04	324.6	13.77	316.1	4.444e+07

Objective Convergence



Constraint Convergence

